Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
enkorol@enkorol:~$ ls /etc/ > file.txt
enkorol@enkorol:~$ ls >> file.txt
enkorol@enkorol:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
```

Рисунок 1: Запись в файл

Работа фильтра

```
enkorol@enkorol:~$ grep .conf file.txt > conf.txt
enkorol@enkorol:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/enkorol/.npm/_cacache/index-v5/1a/ce
/home/enkorol/.npm/_cacache/index-v5/c4
/home/enkorol/.npm/_cacache/index-v5/1d/f1/ce5e4168fa5f662c23cbbd558c5d82ef5e57d137e2d329fbc17126408620
/home/enkorol/.npm/_cacache/index-v5/68/ca
/home/enkorol/.npm/_cacache/index-v5/9e/e0/c3386deeef310820ca85762b73df78a4cca889774c30364b036d1f870889
/home/enkorol/.npm/_cacache/index-v5/c7
/home/enkorol/.npm/_cacache/index-v5/c7/b3/c81fcd46b7c6e984952f72b5d5823025eb825b31eb9b584107047503dd5a
/home/enkorol/.npm/_cacache/index-v5/f3/6c/c3986891a2eed646eaa7d914d91cffc8d6421272c959bdb5a87486645945
/home/enkorol/.npm/_cacache/index-v5/84/ce
/home/enkorol/.npm/_cacache/index-v5/09/c0
/home/enkorol/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc29def30220242c53
/home/enkorol/.npm/_cacache/index-v5/ce
/home/enkorol/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad55fb2000f59e2edcc
/home/enkorol/.npm/_cacache/index-v5/cd
/home/enkorol/.npm/_cacache/index-v5/c9
/home/enkorol/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea310cda0b203217f693
/home/enkorol/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d1350482a1d990504757
/home/enkorol/.npm/_cacache/index-v5/c0
/home/enkorol/.npm/_cacache/index-v5/af/cf
/home/enkorol/.npm/_cacache/index-v5/c3
/home/enkorol/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f2506214f94db8861160d3
/home/enkorol/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c8188a44d554f3f3ba2e
/home/enkorol/.npm/_cacache/index-v5/ca
/home/enkorol/snap/hugo/common
/home/enkorol/snap/hugo/current
/home/enkorol/conf.txt
enkorol@enkorol:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/brltty/Text/hy.ttb
/etc/containers/oci/hooks.d
/etc/foomatic/hashes.d
/etc/foomatic/hashes.d/hashes.new
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
/etc/logrotate.d/httpd
'/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sssd': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
enkorol@enkorol:~$ find ~ -name "log*" > logfile &
[1] 7181
enkorol@enkorol:~$
[1]+  Done                    find ~ -name "log*" > logfile
enkorol@enkorol:~$ rm logfile
enkorol@enkorol:~$
enkorol@enkorol:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
enkorol@enkorol:~$
enkorol@enkorol:~$ gedit &
[1] 7225
enkorol@enkorol:~$ ps | grep gedit
   7225 pts/0    00:00:00 gedit
enkorol@enkorol:~$ kill 7225
[1]+  Terminated              gedit
enkorol@enkorol:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This manual page documents the GNU version of df.  df displays the amount of space available on
       the file system containing each file name argument.  If no file name is given, the space avail-
       able  on  all currently mounted file systems is shown.  Space is shown in 1K blocks by default,
       unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

       If an argument is the absolute file name of a device node containing a mounted file system,  df
       shows the space available on that file system rather than on the file system containing the de-
       vice  node.   This version of df cannot show the space available on unmounted file systems, be-
       cause on most kinds of systems doing so requires non-portable intimate knowledge of file system
       structures.

OPTIONS
       Show information about the file system on which each FILE resides, or all file systems  by  de-
       fault.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576
 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent  sizes  rather  than device usage; although the apparent size is usually
              smaller, it may be larger due to holes in ('sparse') files, internal fragmentation,  in-
              direct blocks, and the like

       -B, --block-size=SIZE
              scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576
              bytes; see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'
 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
enkorol@enkorol:~$
enkorol@enkorol:~$ df
Filesystem     1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 90603152  63944336  59% /
devtmpfs         4011272       0   4011272   0% /dev
tmpfs            4035644     100   4035544   1% /dev/shm
tmpfs            1614260    1944   1612316   1% /run
tmpfs               1024       0      1024   0% /run/credentials/systemd-journald.service
tmpfs            4035648    1612   4034036   1% /tmp
/dev/nvme0n1p3 155186176 90603152  63944336  59% /home
/dev/nvme0n1p2   1992552  518800   1352512  28% /boot
/dev/loop1        115712  115712         0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop0         68480   68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2         49280   49280         0 100% /var/lib/snapd/snap/snapd/25935
tmpfs               1024       0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807128     180    806948   1% /run/user/1053
/dev/sr0         2677920 2677920         0 100% /run/media/enkorol/Fedora-WS-Live-43
enkorol@enkorol:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4        ./.npm/_cacache/index-v5/aa
4        ./.npm/_cacache/index-v5/30/1a
4        ./.npm/_cacache/index-v5/30
424      ./.npm/_cacache/index-v5
115644   ./.npm/_cacache
115708   ./.npm
16       ./snap/hugo/25676/.config/go/telemetry/local
0        ./snap/hugo/25676/.config/go/telemetry/upload
16       ./snap/hugo/25676/.config/go/telemetry
16       ./snap/hugo/25676/.config/go
16       ./snap/hugo/25676/.config
16       ./snap/hugo/25676
0        ./snap/hugo/common
20       ./snap/hugo
20       ./snap
0        ./monthly
0        ./reports/monthly/monthly
0        ./reports/monthly
0        ./reports
4        ./ski.plases/equipment
0        ./ski.plases/plans
4        ./ski.plases
0        ./australia
0        ./play/games/play
0        ./play/games
0        ./play
1521664  .
enkorol@enkorol:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/enkorol/.npm/_cacache/index-v5/aa/bf
/home/enkorol/.npm/_cacache/index-v5/30
/home/enkorol/.npm/_cacache/index-v5/30/1a
/home/enkorol/snap
/home/enkorol/snap/hugo
/home/enkorol/snap/hugo/25676
/home/enkorol/snap/hugo/25676/.config
/home/enkorol/snap/hugo/25676/.config/go
/home/enkorol/snap/hugo/25676/.config/go/telemetry
/home/enkorol/snap/hugo/25676/.config/go/telemetry/local
/home/enkorol/snap/hugo/25676/.config/go/telemetry/upload
/home/enkorol/snap/hugo/common
/home/enkorol/monthly
/home/enkorol/reports
/home/enkorol/reports/monthly
/home/enkorol/reports/monthly/monthly
/home/enkorol/ski.plases
/home/enkorol/ski.plases/equipment
/home/enkorol/ski.plases/plans
/home/enkorol/australia
/home/enkorol/play
/home/enkorol/play/games
/home/enkorol/play/games/play
enkorol@enkorol:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.